



**NANYANG
TECHNOLOGICAL
UNIVERSITY**
SINGAPORE

**College of Computing and
Data Science**

SC2104/CE3002: Sensors, Interfacing and Digital Control

Topic 6 – Kalman Filter

Assoc Prof Nicholas Vun

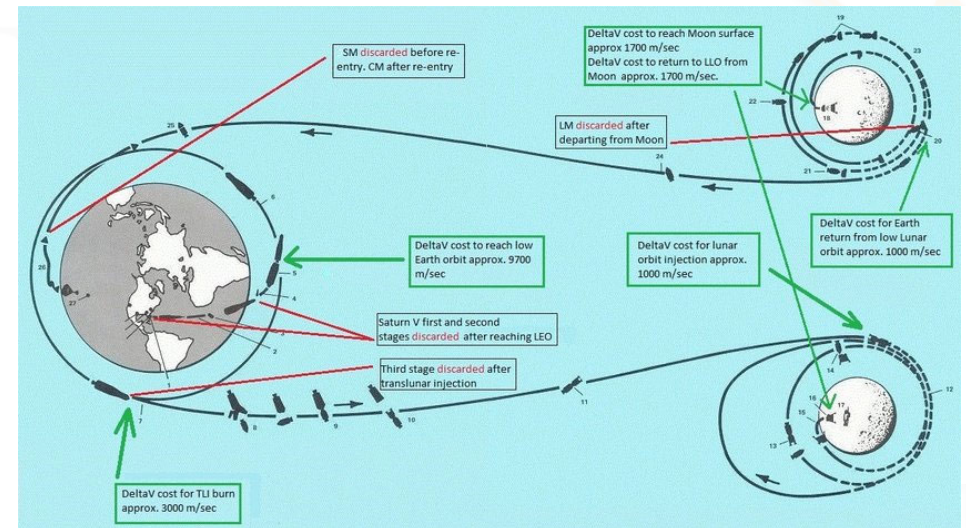
Introduction

Kalman filter is an algorithm that are commonly used to fuse multiple measurements/information in real time

- to provide accurate prediction of the state of a system (e.g., the roll and pitch angles of an IMU)

Invented by Rudolf Emil Kálmán

- most widely known application: used on Apollo lunar missions to give precise estimates of the spacecrafts position
- enable the complex orbital trajectories essential to reach the moon

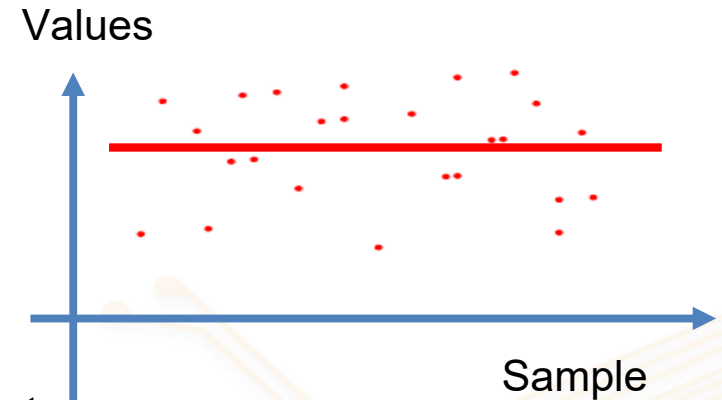


Source: forum.nasaspaceflight.com

Kalman Filter vs Other Estimations

Some common approaches to estimate the true value

- (i) Averaging
- (ii) Least Square - minimize the difference between (collected) measurements and true value
- (iii) Recursive Least Square – Least square performed in real time with stream of data to update the estimate of a static parameter



Kalman Filter

- recursively update and estimate the evolving state
- i.e., the state changes over time (E.g., the location of an autonomous vehicle travelling on the road)
- using appropriate **state model** and **measurement model**

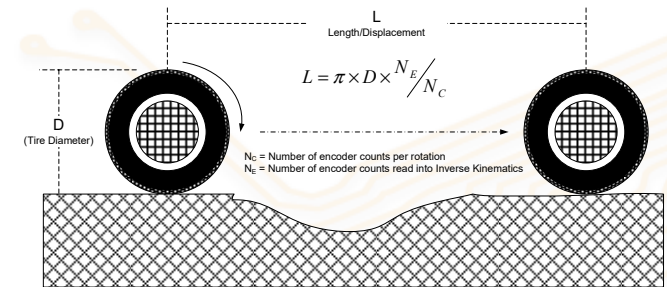
Estimation of position

An autonomous car driving along the road

- how do we find (estimate) its position?

1) Use the Odometer - based on number of turns of the wheels

- Depends on tire diameter
- Tire diameter is dependent on:
 - Temperature, tread wear, air pressure
- Tire slip
- Non-uniform surface etc.



Bottom line:

With well-calibrated tires, the odometry-based “dead reckoning” measurement can only provide useful position information over short distances (e.g., up to 20 meters)

Estimation of position

2a) State modelling using certain process model

Distance travel = speed x time

(e.g., use speedometer in the car to measure speed)

position
$$\mathbf{x} = \mathbf{x}_0 + \dot{\mathbf{x}} * \Delta t$$

where $\dot{\mathbf{x}}$ = speed and $\mathbf{x}_0$ = original position.

This is usually expressed in a more general form:

$$\mathbf{x}_k = \mathbf{x}_{k-1} + \dot{\mathbf{x}}_{k-1} * \Delta t$$

where $\mathbf{x}_k$ and $\dot{\mathbf{x}}_k$ are the current position and speed (at time instant k), and $\mathbf{x}_{k-1}$ is the previous position (at time instant $k - 1$).

- these are known as the **state** of the system

Estimation of position

2b) State model – if there is also a constant acceleration $\ddot{x}_k$ during the time interval (measured using accelerometer)

position $x_k = x_{k-1} + \dot{x}_{k-1} * \Delta t + \frac{1}{2} * \ddot{x}_k * \Delta t^2$

velocity $\dot{x}_k = \dot{x}_{k-1} + \ddot{x} * \Delta t$

Aside: We usually represent such equations using matrices to combine them to form one state matrix equation

$$\begin{bmatrix} x_k \\ \dot{x}_k \end{bmatrix} = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_{k-1} \\ \dot{x}_{k-1} \end{bmatrix} + \begin{bmatrix} \frac{1}{2}\Delta t^2 \\ \Delta t \end{bmatrix} [\ddot{x}_k]$$

This is more convenient because

- (i) In practice, we will have more than just 2 equations
E.g., we can have up to 6 such equations in the case of 3 dimensions in the real world (i.e. x, y and z)
- (ii) Matrices manipulation functions are available in software libraries that allow matrices to be solved very efficiently using computer

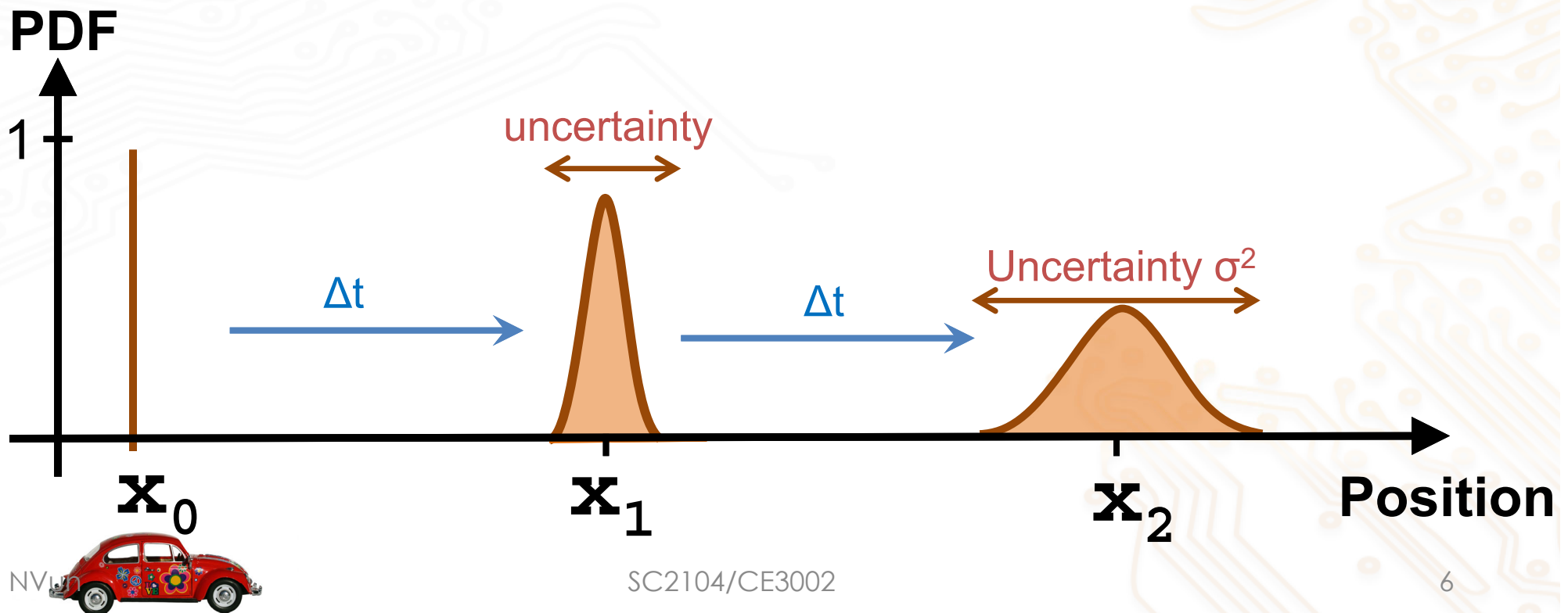
Uncertainty (Error)

Consider the case of a vehicle moving in the x direction

- uncertainty occurs in the position estimated using the state process equation $\mathbf{x} = \mathbf{x}_0 + \dot{\mathbf{x}} * \Delta t$
- uncertainty will grow as it moves forward

Uncertainty is described by its **variance σ^2**

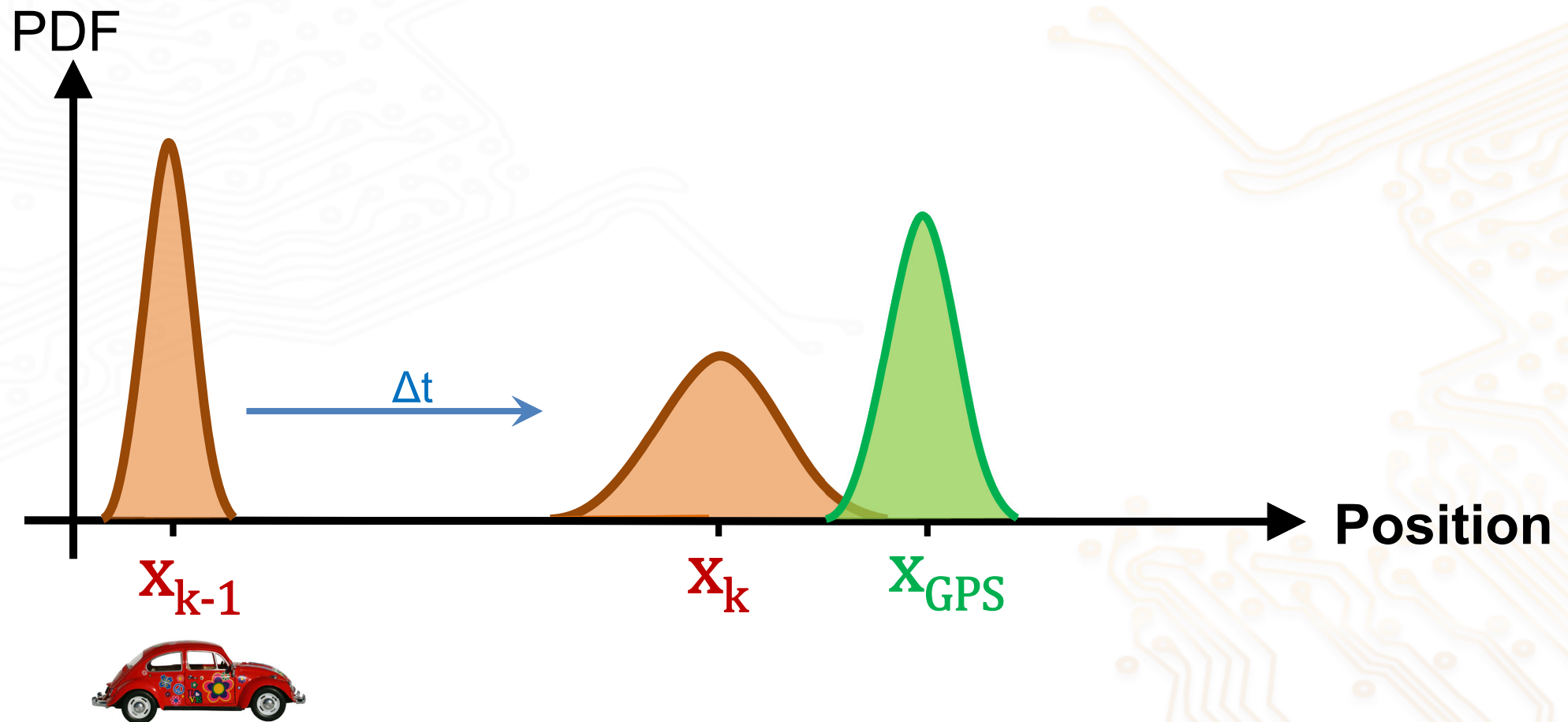
- which measure the spread of the possible position



Fusion - Intuition

Assuming that we have another measurement that can be used to provide the position of the vehicle

- e.g., GPS signal (which has its own uncertainty)



Fusion - Intuition

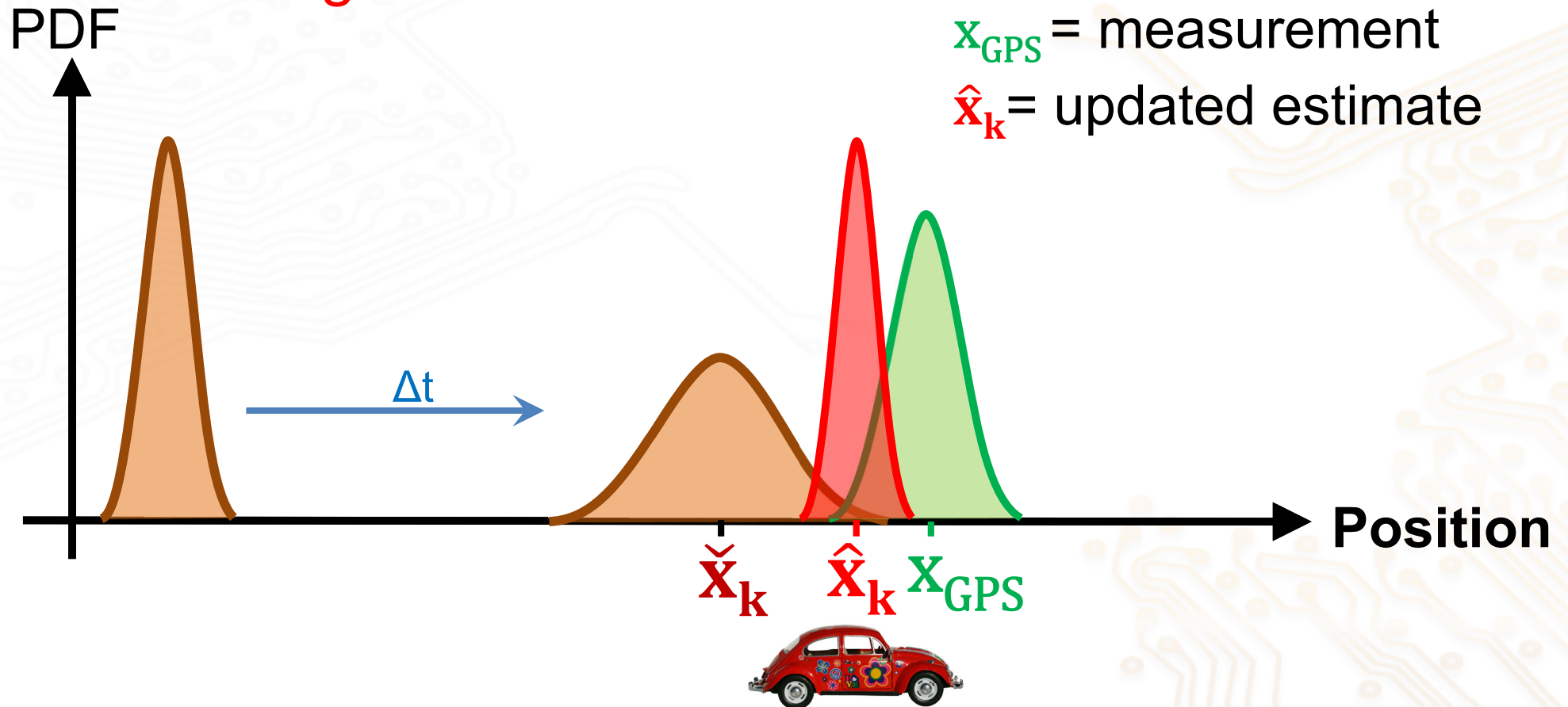
We can better estimate the position of the vehicle

- by taking into consideration both values and their uncertainties
- i.e. **fusing** the measurements

$\check{\mathbf{x}}_k$ = process estimate

$\mathbf{x}_{GPS}$ = measurement

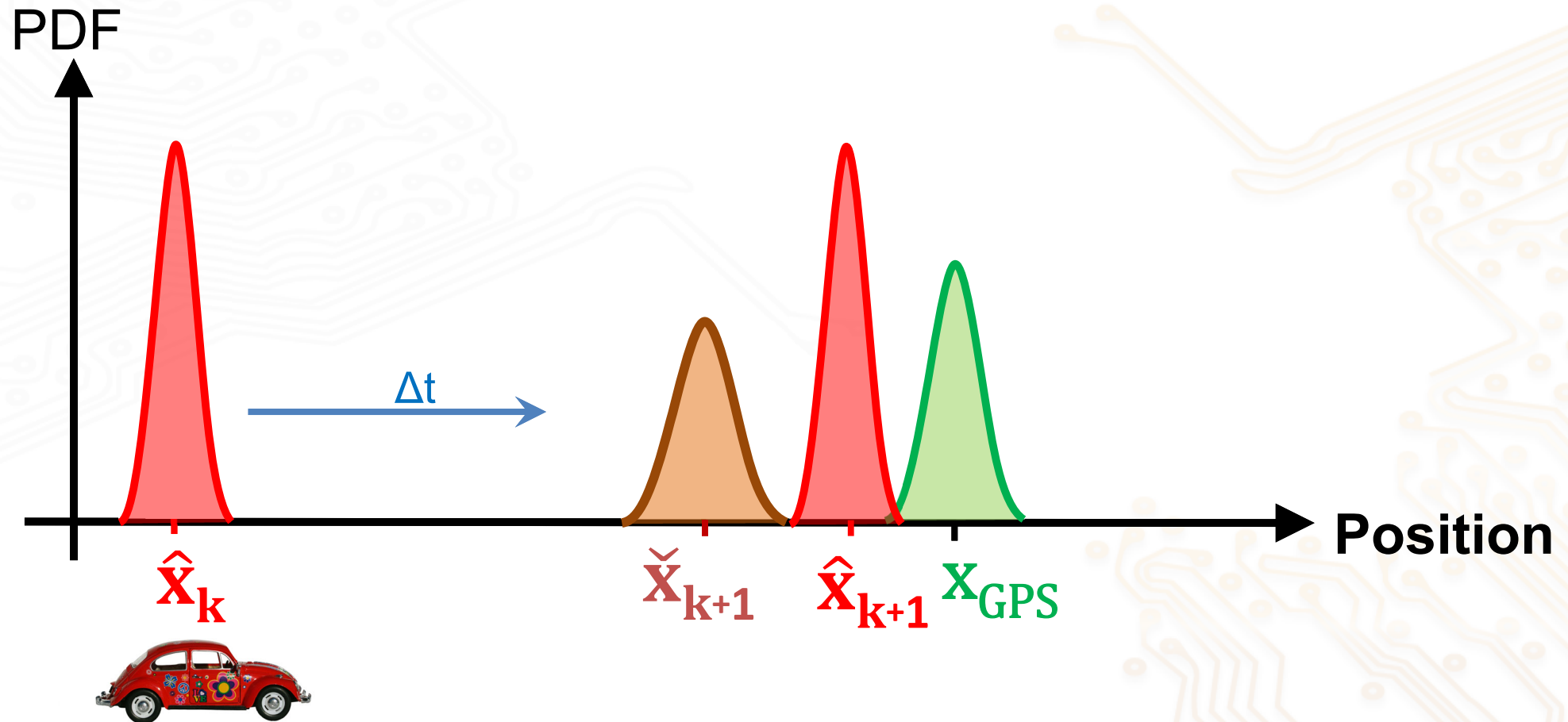
$\hat{\mathbf{x}}_k$ = updated estimate



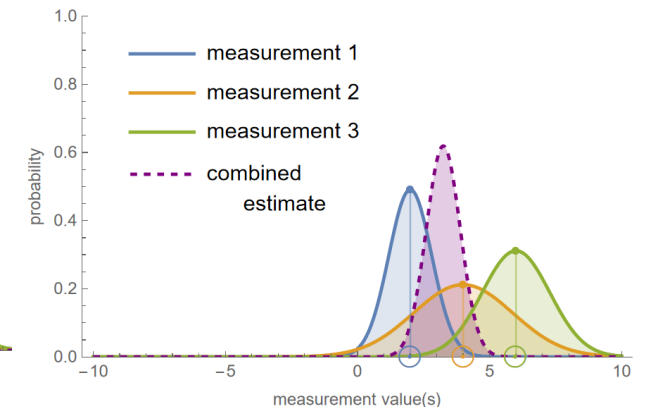
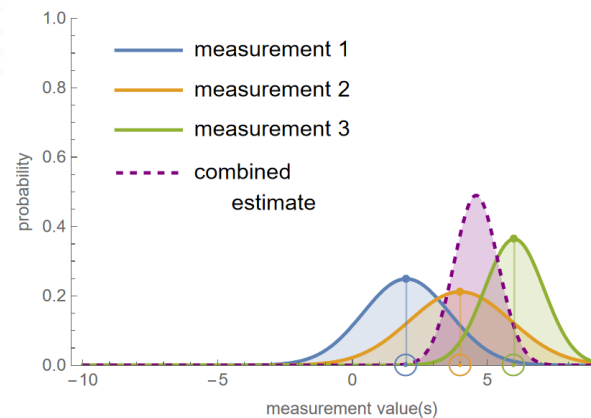
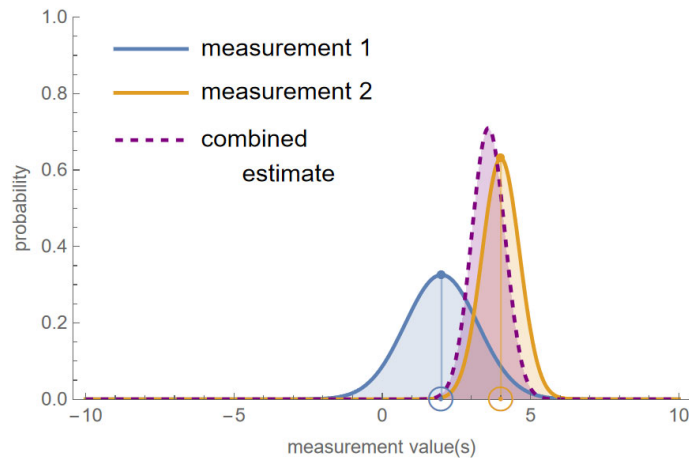
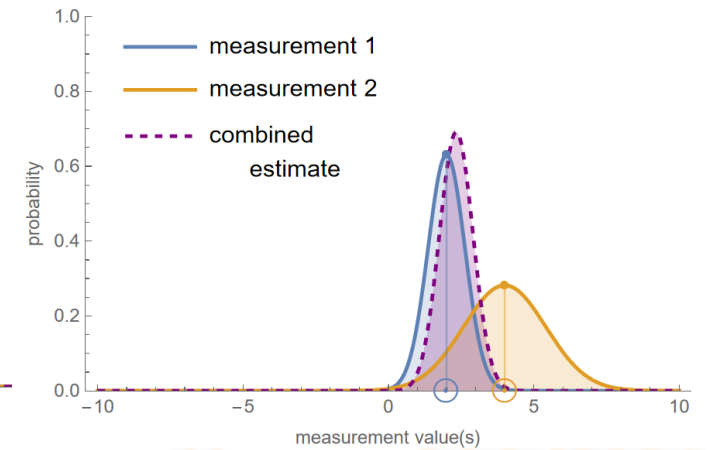
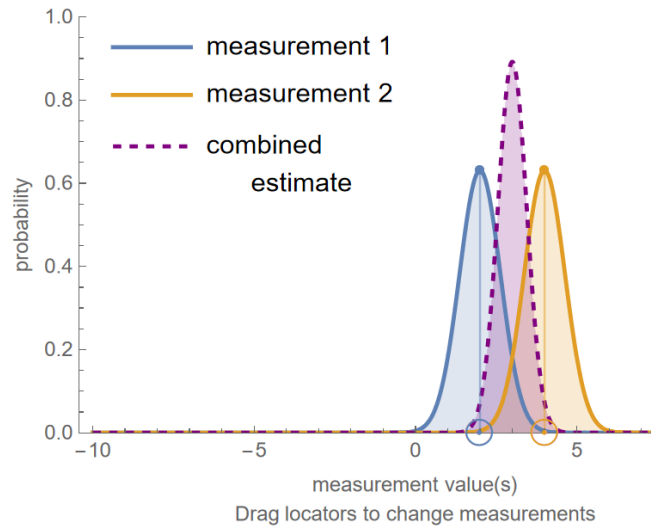
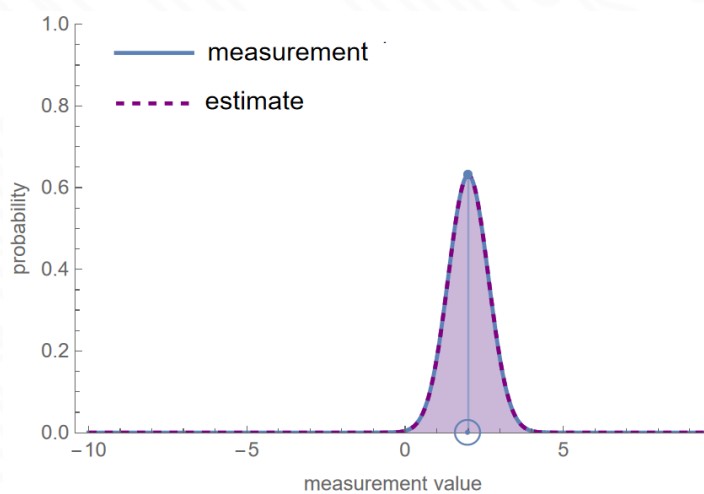
Fusion - Intuition

The corrected estimate $\hat{\mathbf{x}}_k$ is used by the process to compute its next estimate position $\check{\mathbf{x}}_{k+1}$

- and recursively corrected by using fusion with the measurement given by the second sensor for $\hat{\mathbf{x}}_{k+1}$



Fusion (and Baynes Theorem)



$$\hat{x}_{12} = \bar{x}_1 \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2} + \bar{x}_2 \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2}$$

$$\hat{\sigma}_{12}^2 = \frac{\sigma_1^2 \sigma_2^2}{\sigma_1^2 + \sigma_2^2}$$

Kalman Filter

An algorithm that use the principle of fusion to predict the state of the system

- using a **Kalman Gain equation** to continuously update the estimate value and its variance

Recursive application of the **Kalman Gain K_G** to the process state equation and measurement

- continuously reduce the uncertainty of each round of estimate
- lead to more precise estimate

Aside: It is called a “Filter” as it reduces the uncertainty (i.e., noise) of the state of the system

Kalman Filter Algorithm

The Kalman Filter consists of the following stages:

0. a) Establish the uncertainty of the model process and measurement uncertainty
 - provided by the sensor's vendor, calculated, or derived empirically by a calibration procedure.
- b) Estimate the initial states and its uncertainty
1. Computed the **model process** state estimate
e.g. $\check{\mathbf{x}}_k = \hat{\mathbf{x}}_{k-1} + \dot{\mathbf{x}} * t$
or pitch and roll angles from **Gyroscope**
2. Calculate the uncertainty of the process estimate $\check{\sigma}_k^2$
3. Calculate the Kalman Gain K_G (using the updated uncertainty of the estimate $\check{\sigma}_k^2$ and uncertainty of measurement established in 0)

Steps for Kalman Filter Algorithm

1. Computed the model **process** state estimate $\check{x}_k$ (e.g. pitch and roll angles from **Gyroscope**)
2. Calculate the uncertainty of the process estimate $\check{\sigma}_k^2$
3. Calculate the Kalman Gain K_G (using the updated uncertainty of the estimate $\check{\sigma}_k^2$ and measurement uncertainty established in 0)
4. Get the **measurement** value by sensor (e.g. pitch and roll angles from **Accelerometer**)
5. Use K_G , model estimate $\check{x}_k$ and sensor measurement to calculate the corrected estimate $\hat{x}_k$
6. Calculate the uncertainty of the corrected estimate $\hat{\sigma}_k^2$ (which will become the process uncertainty to be updated at time $K+1$, i.e. $\check{\sigma}_{k+1}^2$ in 2 above)
7. Repeat from 1

Kalman Filter Algorithm for IMU

0. a) Established the tolerance/variance of the process and the tolerance/variance of measuring sensor

Based on standard deviation of Gyroscope accuracy = 4 and Accelerometer accuracy = 3:

- Variance of Gyroscope estimate $\text{Var}_G = 4^2$
- Variance of Accelerator measurement $\text{Var}_A = 3^2$

(Note: These values do not change as the state evolves.)

- b) Estimate the initial state and its uncertainty (of Gyro)

Assume 0 degree but the gyro angles may not be at 0 at the beginning e.g. gyro is not lying flat

- $\text{roll}_g = 0$ degree with uncertainty $\text{roll}_g\text{var} = 2^2$
- $\text{pitch}_g = 0$ degree with $\text{pitch}_g\text{var} = 2^2$

(Note: These values will change and be updated as the state evolves)

Kalman Filter Algorithm for IMU

1. Obtain the measurement values by sensor
i.e. pitch and roll angles from Accelerometer

Using the approximate* roll and pitch angles
formulae for accelerometer

$$\text{roll_a} = \tan^{-1} \left[\frac{a_x}{a_z} \right]$$

$$\text{pitch_a} = \tan^{-1} \left[\frac{a_y}{a_z} \right]$$

* Approximate because we assume that the IMU is either
been roll or pitch, but not both at the same time

Kalman Filter Algorithm for IMU

2. Computed the process state estimate $\check{\mathbf{x}}_k$

i.e. get the values from **Gyroscope** and calculate the estimates

$$\text{roll_g} = \text{roll_g} + \Delta t * \text{gyro_x}$$

$$\text{pitch_g} = \text{pitch_g} + \Delta t * \text{gyro_y}$$

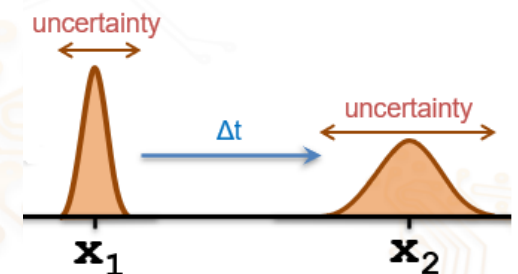
3. Calculate the uncertainty of the process estimate $\check{\sigma}_k^2$

As the roll and pitch angles are obtained by 'integration' through multiplication: $\Delta t * \text{gyro_value}$

- it can be shown that the variance will change according to the square of Δt

$$\text{roll_g_var} = \text{roll_g_var} + \Delta t^2 * \text{Var}_G$$

$$\text{pitch_g_var} = \text{pitch_var_g} + \Delta t^2 * \text{Var}_G$$



Kalman Filter Algorithm for IMU

4. Calculate the Kalman Gain K_G (using the updated uncertainty of the process estimate and uncertainty of measurement sensor)

Kalman Gain K_G equation:
$$K_G = \frac{\text{Var}_{\text{est}}}{\text{Var}_{\text{est}} + \text{Var}_{\text{mea}}}$$

i.e.
$$K_{G_{\text{roll}}} = \frac{\text{roll_g_var}}{\text{roll_g_var} + \text{Var}_A}$$

$$K_{G_{\text{pitch}}} = \frac{\text{pitch_g_var}}{\text{pitch_g_var} + \text{Var}_A}$$

Note: Range of K_G will vary between 0 and 1

$K_G \rightarrow 1$ if $\text{Var}_{\text{est}} \gg \text{Var}_{\text{mea}}$: i.e. measured value is more accurate

$K_G \rightarrow 0$ If $\text{Var}_{\text{est}} \ll \text{Var}_{\text{mea}}$: i.e. process value is more accurate

Kalman Filter Algorithm for IMU

5. Use $\mathbf{K}_G$, process estimate $\tilde{\mathbf{x}}_k$ and sensor measured value to calculate the corrected estimate $\hat{\mathbf{x}}_k$

a. The estimates are corrected by using the formulae

$$\text{roll_KF} = \text{roll_g} + K_{G_{\text{roll}}}(\text{roll_a} - \text{roll_g})$$

$$\text{pitch_KF} = \text{pitch_g} + K_{G_{\text{pitch}}}(\text{pitch_a} - \text{pitch_g})$$

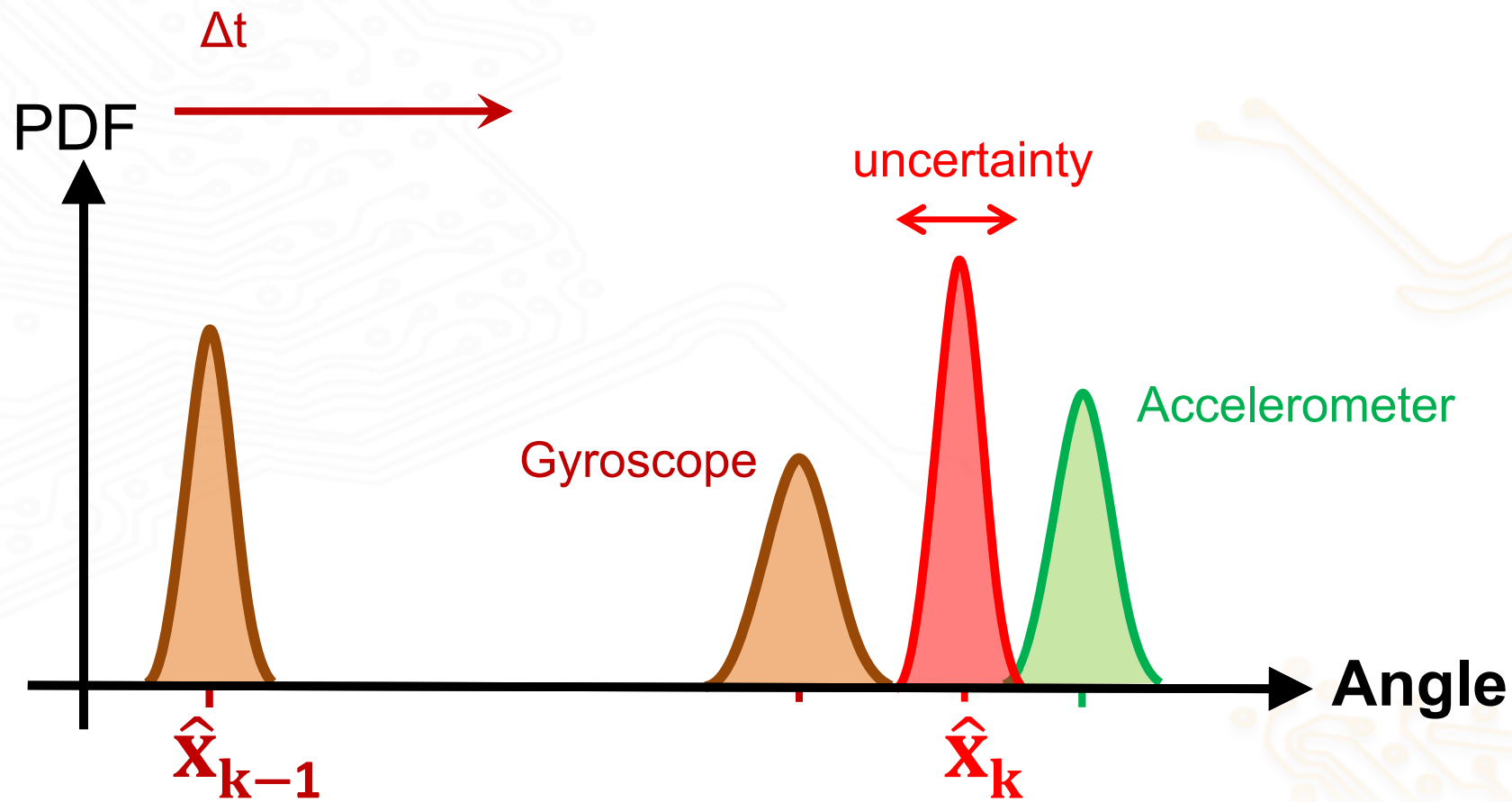
b. Update the estimates (for next iteration):

$$\text{roll_g} = \text{roll_KF} \quad \text{and} \quad \text{pitch_g} = \text{pitch_KF}$$

Note:

If $K_G \rightarrow 0$ (i.e. $\text{Var}_{\text{est}} \ll \text{Var}_{\text{mea}}$), the corrected estimate will be based largely on gyroscope.

If $K_G \rightarrow 1$ (i.e. $\text{Var}_{\text{est}} \gg \text{Var}_{\text{mea}}$), the corrected estimate will be based largely on accelerometer.



Kalman Filter Algorithm for IMU

6. Calculate the uncertainty of the corrected estimate

- a. The uncertainty of the corrected estimate is obtained by using the equation

$$\hat{\sigma}^2 = (1 - K_G)\check{\sigma}^2$$

$$\text{roll_var} = (1 - K_G) \text{roll_g_var}$$

$$\text{pitch_var} = (1 - K_G) \text{pitch_g_var}$$

- b. These are used as the uncertainties of the gyro estimates in subsequent iteration of computation

$$\text{roll_g_var} = \text{roll_var}$$

$$\text{pitch_g_var} = \text{pitch_var}$$

7. Repeat from 1

Summary

Kalman filter is an algorithm that operates based on fusion principle

- uses a parameter, the Kalman Gain to derive an optimum estimate of the state of the system

By comparing the variance (uncertainty) of the information from multiple sources

- it recursively adjusts the estimation as the state evolves to minimize the variance
- fusing the information to achieve optimum prediction

Aside: 1. Kalman Filter only works with systems whose variance is gaussian

2. Extended Kalman Filter is used for non-linear system